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Roll No:-2301028
AI and DS
Division:T1
Batch:T13
Subject: ASD & DevOps

EXPERIMENT:-1

Aim: Study and Implementation of Basic Git and GitHub Operations.

This Experiment maps to LO2.

LO2: To understand and perform the basic version control operations using Git and GitHub, including repository creation, cloning, file tracking, committing, pushing, pulling, branching, merging, and remote repository management.

Theory: Git is a distributed version control system that helps developers track changes in source code and files. It enables efficient project management through repositories, commits, branches, and remote repositories, making collaboration and version management easier. In this experiment, various Git commands are performed to understand the basic operations of version control.

Commands:

No.	Command	Purpose
1	git --version	Checks the installed Git version.
2	git config --global user.name "Manya Parwani"	Sets the Git username globally.
3	git config --global user.email "email"	Sets the Git email globally.
4	git config --list	Displays Git configuration details.
5	mkdir gitproject	Creates a new project folder.
6	cd gitproject	Enters the project folder.
7	git init	Initializes a new Git repository.
8	echo "Hello Git" > file.txt	Creates a new file with content.
9	cat file.txt	Displays the contents of the file.
10	git status	Checks the current repository status.

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11	git add file.txt	Adds a specific file to the staging area.
12	git add .	Adds all modified files to the staging area.
13	git commit -m "Initial commit"	Creates the first committed version.
14	git log	Displays detailed commit/version history.
15	git log --oneline	Displays different versions using short commit IDs.
16	echo "Git Version Control" >> file.txt	Adds content to an existing file.
17	git diff	Displays changes between the current and committed file.
18	git restore file.txt	Rolls back uncommitted file changes.

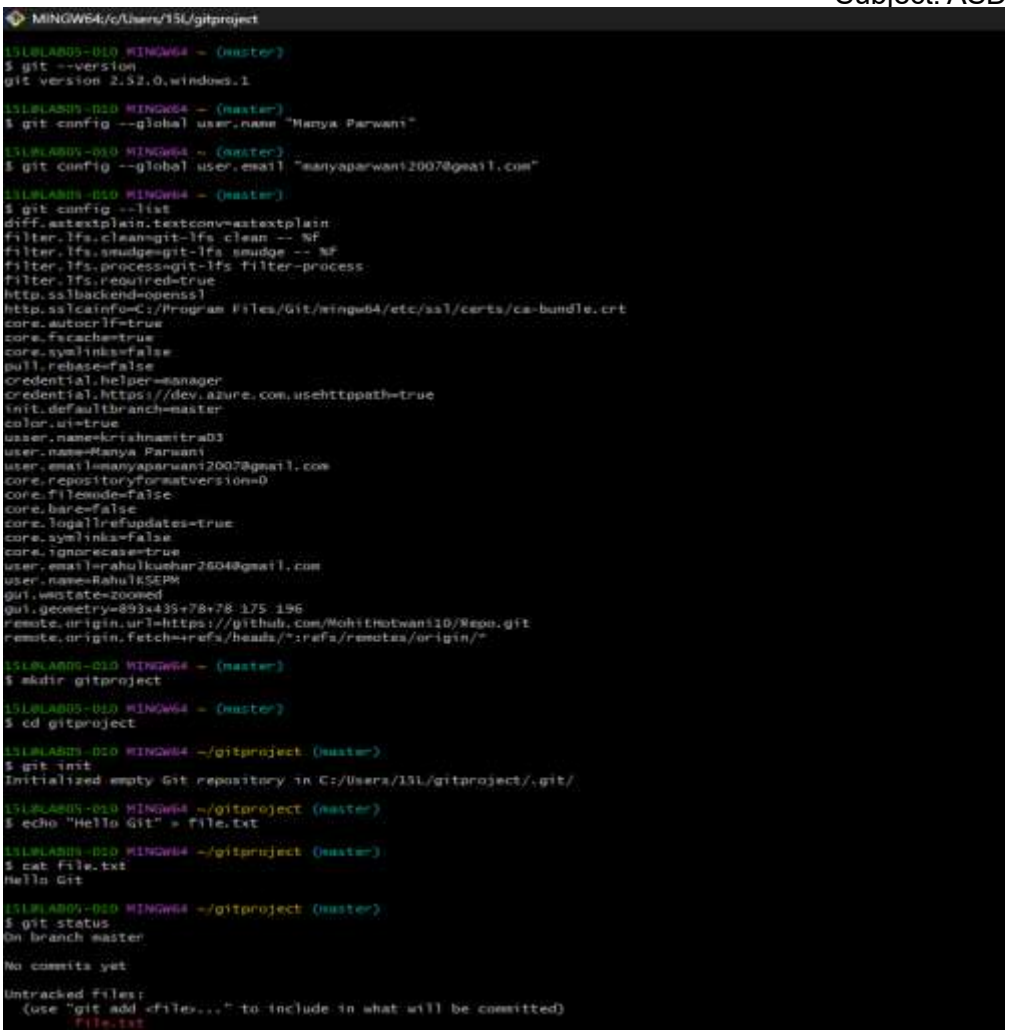
19	git reset HEAD~1	Rolls back the latest commit while retaining changes.
20	git branch feature	Creates a new branch named feature.
21	git branch	Displays all branches and the current branch.
22	git checkout feature	Switches to the feature branch.
23	git switch feature	Switches to the feature branch using the modern Git command.
24	echo "Feature Branch" > feature.txt	Creates a file in the feature branch.
25	git checkout main	Switches back to the main branch.
26	git switch main	Switches back to the main branch using git switch.
27	git merge feature	Merges the feature branch with the current branch.
28	git branch -d feature	Deletes the feature branch.
29	git remote add origin <URL>	Connects the local repository to a remote repository.
30	git remote -v	Displays configured remote repositories.
31	git push -u origin main	Pushes local commits to the remote repository and authenticates if required.

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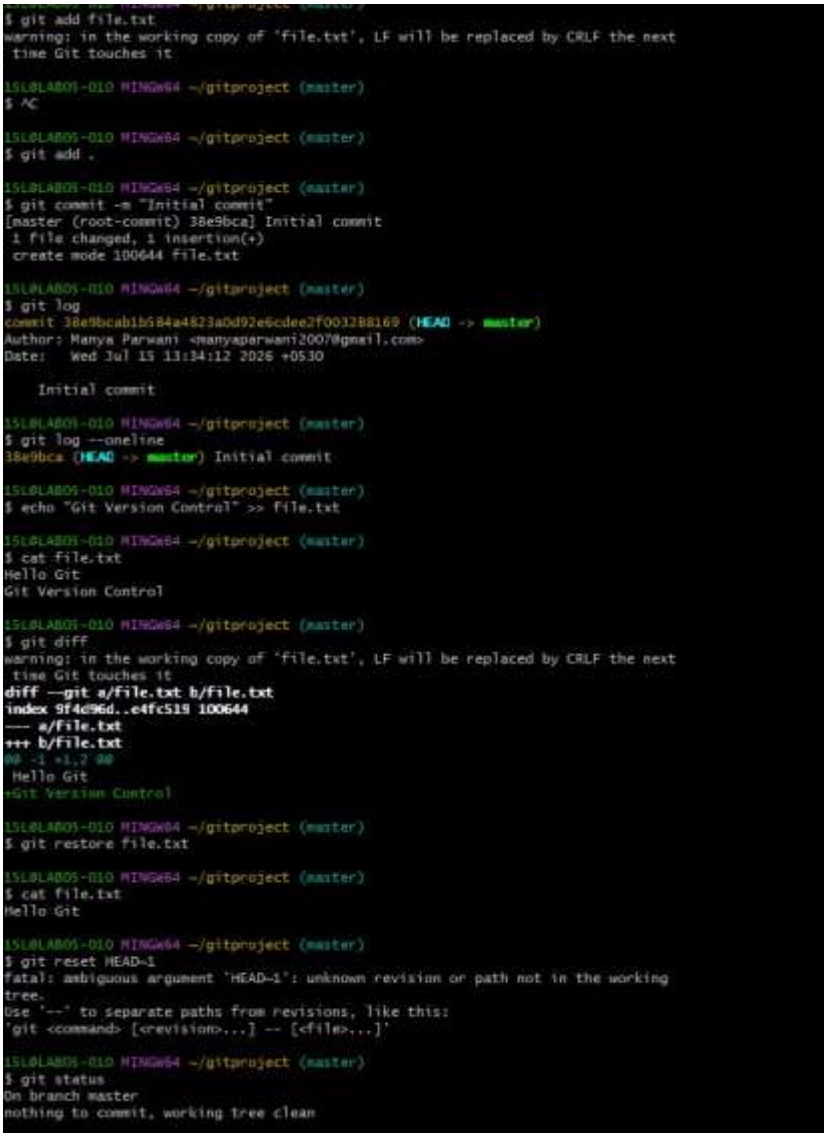
32	git pull origin main	Retrieves and merges changes from the remote repository.
33	git clone <URL>	Creates a local copy of a remote repository.
34	git show <commit-id>:file.txt	Checks the content of a file in a specific version.
35	git checkout <commit-id>	Switches the repository to a previous committed version.
36	git checkout main	Returns from an older version to the latest main branch.
37	git restore --source=<commit-id> file.txt	Restores a file from a specific older version.

Output:

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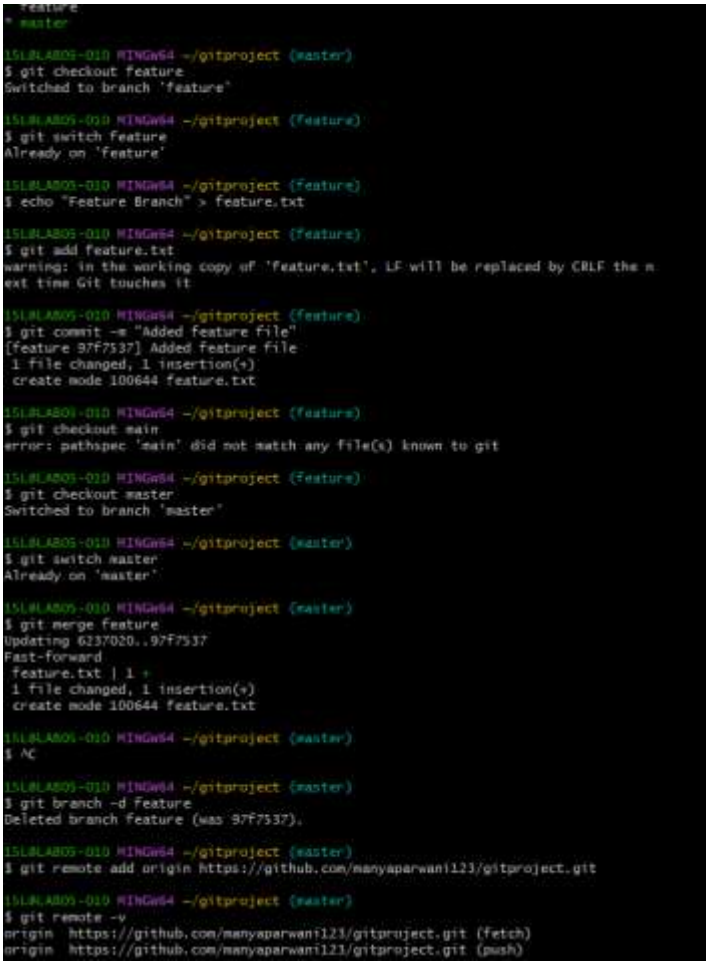
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```
15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git push -u origin master
branch 'master' set up to track 'origin/master'.
Everything up-to-date

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ ^C

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git pull origin master
From https://github.com/manyaparwani123/gitproject
 * branch            master     -> FETCH_HEAD
Already up to date.

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ cd ..

15L@LAB05-010 MINGW64 ~ (master)
$ git clone https://github.com/manyaparwani123/gitproject.git
fatal: destination path 'gitproject' already exists and is not an empty director
y.

15L@LAB05-010 MINGW64 ~ (master)
$ ^C

15L@LAB05-010 MINGW64 ~ (master)
$ git clone https://github.com/manyaparwani123/gitproject.git gitproject_clone
Cloning into 'gitproject_clone'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 9 (delta 0), reused 9 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
```

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Conclusion:

The experiment was successfully completed using basic Git commands to create and manage a repository, track changes, perform commits, manage branches, and work with a remote GitHub repository. It helped in understanding the fundamentals of Git and version control.